

Started on	Thursday, 30 April 2026, 2:46 PM
State	Finished
Completed on	Thursday, 30 April 2026, 2:50 PM
Time taken	3 mins 22 secs
Grade	100.00 out of 100.00

Question 1

Correct

Mark 100.00 out of 100.00

Write a program to find the eigenvalues and associated eigenvectors for the matrix

$\begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

For example:

Result

Eigen values are [1. -4. 3.] and Eigen Vectors are $\begin{bmatrix} 0.9486833 & 0.4472136 & 0. \\ 0.31622777 & 0.89442719 & 0. \\ 0. & 0. & 1. \end{bmatrix}$

Answer: (penalty regime: 0 %)

Reset answer

```
1 #Program to find the eigen values and eigen vectors.
2 #Developed by:DINESH D
3 #RegisterNumber:212225040079
4 import os
5 os.environ["OPENBLAS_NUM_THREADS"]="1"
6 import numpy as np
7 matrixA = np.array([[2,-3,0],[2,-5,0],[0,0,3]])
8 eigenvalues, eigenvectors = np.linalg.eig(matrixA)
9 print("Eigen values are", eigenvalues, "and Eigen Vectors are", eigenvectors)
```

	Expected	Got	
✓	Eigen values are [1. -4. 3.] and Eigen Vectors are $\begin{bmatrix} 0.9486833 & 0.4472136 & 0. \\ 0.31622777 & 0.89442719 & 0. \\ 0. & 0. & 1. \end{bmatrix}$	Eigen values are [1. -4. 3.] and Eigen Vectors are $\begin{bmatrix} 0.9486833 & 0.4472136 & 0. \\ 0.31622777 & 0.89442719 & 0. \\ 0. & 0. & 1. \end{bmatrix}$	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 100.00/100.00.

◀ EX-03-DISCUSSION

Jump to...

Exp-04 - Record Eigen Values and Eigen Vectors ►